Managerial Economics

Lecture 3 Quantitative Demand Analysis

Zhiyuan Chen

iMBA, Fall Semester 2025
@Tongzhou Campus

Opening discussion

Bob designs professional PPTs at *pptmaster.com*. Bob charges 200 yuan per file (with 100 pages), and currently polish 12 files per month. His costs are rising (including the opportunity cost of his time), so he considers raising the price to 250 yuan. The law of demand states that he won't sell as many services if he raises his price.



How will Bob's revenue change?

Overview

- Own Price Elasticity of Demand
  - Definition of Demand Elasticity
  - Determinants of Demand Elasticity
  - Variety of Demand Curves
  - Elasticity of Demand and Revenue
- Other Demand Elasticities
  - Cross Price Elasticity of Demand
  - Income Elasticity

Elasticity of Demand

Elasticity（弹性）

- A term borrowed from physics
  - elasticity of spring
  - elastic skin …





Elasticity（弹性）

- **Basic idea:** Elasticity measures how much one variable responds to changes in another variable, *e.g.*
  - Elasticity measures *how much demand for Bob's PPT services will fall if he raises his price*
  - Elasticity measures *how much demand for properties will drop if the housing price drops by 10%*
- **Definition:** **Elasticity** is a numerical measure of the responsiveness of quantity ***Q*** to one of its determinants.

Price Elasticity of Demand

- **Own Price elasticity of demand（需求的价格弹性）** measures how much Q^d responds to a change in P.

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in } Q^d}{\text{1 Percentage change in } P}$$

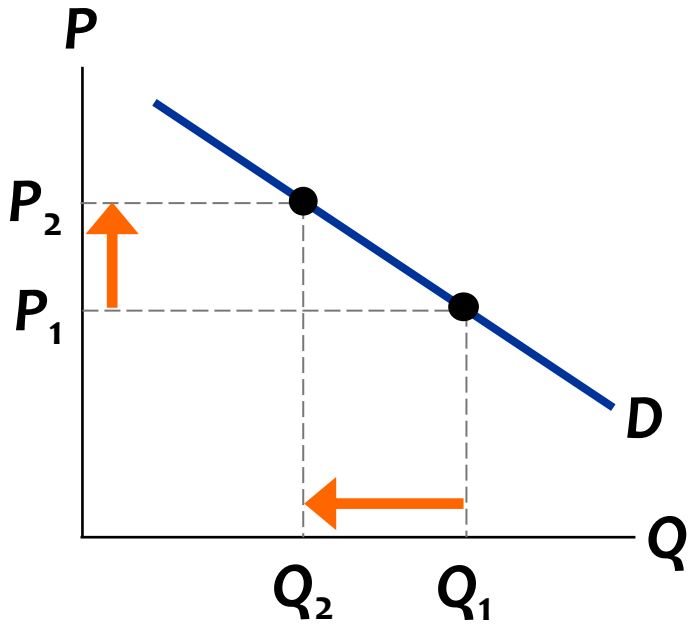
  - We use $E_{Q^d,P}$ to denote the price elasticity of demand
- Loosely speaking, it measures the price-sensitivity of buyers' demand.

Price Elasticity of Demand

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in } Q^d}{\text{1 Percentage change in } P}$$

Example:

P rises by 10%



Q falls by 15%

Price elasticity of demand equals:

$$\frac{-15\%}{10\%} = -1.5$$

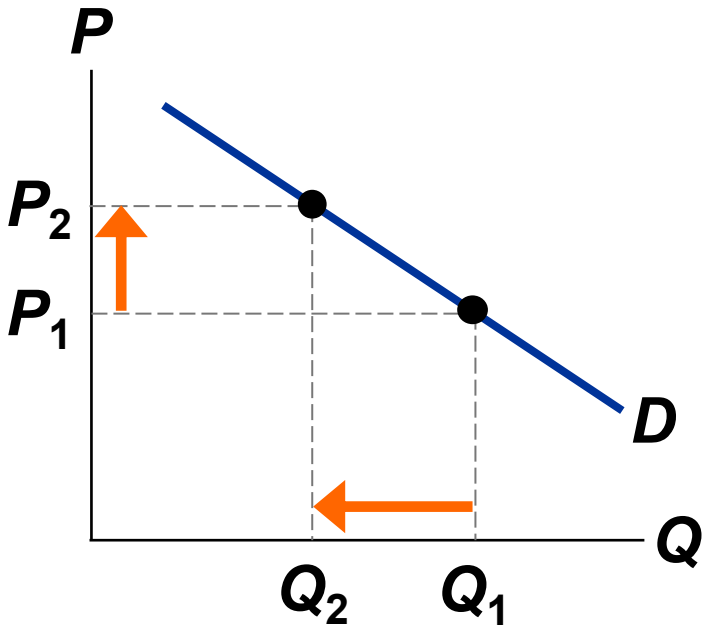
RENMIN BUSINESS SCHOOL

Price Elasticity of Demand

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in } Q^d}{\text{1 Percentage change in } P}$$

Along a ***D*** curve, ***P*** and ***Q*** move in opposite directions, which would make price elasticity negative.

In convention, we drop the minus sign and report all price elasticities as positive numbers.



RENMIN BUSINESS SCHOOL

Calculating Percentage Changes

- We use the **midpoint method（中点法）**:

$$\frac{\text{end value } Q^d - \text{start value } Q^d}{\text{Midpoint of } Q^d} \times 100\%$$

$$\frac{\text{end value } P - \text{start value } P}{\text{Midpoint of } P} \times 100\%$$

- The midpoint is the number halfway between the start & end values, the average of those values

midpoint=(end value + start value)/2

- The bias of this method will be more severe as the gap between end value and start value widens

RENMIN BUSINESS SCHOOL

Calculating Percentage Changes

Using the midpoint method, the % change in *P* equals

$$\frac{\$250 - \$200}{\$225} \times 100\% = 22.2\%$$

The % change in ***Q*** equals

$$\frac{12 - 8}{10} \times 100\% = 40.0\%$$

The price elasticity of demand equals

$$40/22.2 = 1.8$$

Quiz: Calculate an elasticity

Use the following information to calculate the price elasticity of demand for hotel rooms:

if $\boldsymbol{P}$ = \$70, $\boldsymbol{Q}^{\mathbf{d}}$ = 50

if $\boldsymbol{P}$ = \$90, $\boldsymbol{Q}^{\mathbf{d}}$ = 30



Determinants of Demand Elasticity

What Determines Price Elasticity?

To learn the determinants of price elasticity, we look at a series of examples. Each compares two common goods. In each example:

- Suppose the prices of both goods rise by **20%**
- The good for which Q^d falls more (in percent) has the higher price elasticity of demand.

E.G. 1: Breakfast Dumpling vs. Housing

- The prices of both of these goods rise by **20%**. For which good does Q^d drop more? Why?
  - Breakfast dumpling has close substitutes (*e.g.*, cakes, cereal, bread, bun) so buyers can easily switch if the price rises.
  - House has no close substitutes, so consumers would probably not buy much less if its price rises.

Lesson One: ***Price elasticity is higher when close substitutes are available.***

E.G. 2: "Blue Jeans" vs. "Clothing"

- The prices of both goods rise by 20%.
  For which good does Q^d drop the most? Why?
  - For a narrowly defined good such as blue jeans, there are many substitutes (khakis, shorts, trousers).
  - There are fewer substitutes available for broadly defined goods. (There aren't too many substitutes for clothing, *other than living in a nudist colony*.)

Lesson Two: ***Price elasticity is higher for narrowly defined goods than broadly defined ones.***

E.G. 3: Insulin（胰岛素）vs. Rolex Watch

- The prices of both of these goods rise by 20%. For which good does Q^d drop the most? Why?
  - To millions of diabetics（糖尿病人）, insulin is a necessity.
    A rise in its price would cause little or no decrease in demand.
  - A Rolex Watch is a luxury. If the price rises, some people will forego it.

Lesson Three: ***Price elasticity is higher for luxuries than for necessities.***

E.G. 4-Gasoline: Short Run vs. in the Long Run

- The price of gasoline rises 20%. Does Q^d drop more in the short run or the long run? Why?
  - There's not much people can do in the short run, other than ride the bus or carpool.
  - In the long run, people can buy smaller cars or live closer to where they work.

Lesson Four: ***Price elasticity is higher in the long run than the short run.***

Determinants of Price Elasticity: A Summary

The price elasticity of demand depends on:

① *The extent to which close substitutes are available*

② *Whether the good is a necessity or a luxury*

③ *How broadly or narrowly the good is defined*

④ *The time horizon (expectation) – elasticity is higher in the long run than the short run*

Different Types of Demand Curves

The Variety of Demand Curves

- The price elasticity of demand is closely related to the slope of the demand curve.
- **Rule of thumb（拇指法则）：**
  The flatter the curve, the bigger the elasticity.
  The steeper the curve, the smaller the elasticity.

Elastic: $\left|E_{Q^d,P}\right| > 1$

Inelastic: $\left|E_{Q^d,P}\right| < 1$

Unitary: $\left|E_{Q^d,P}\right| = 1$

- Five different classifications of ***D*** curves….

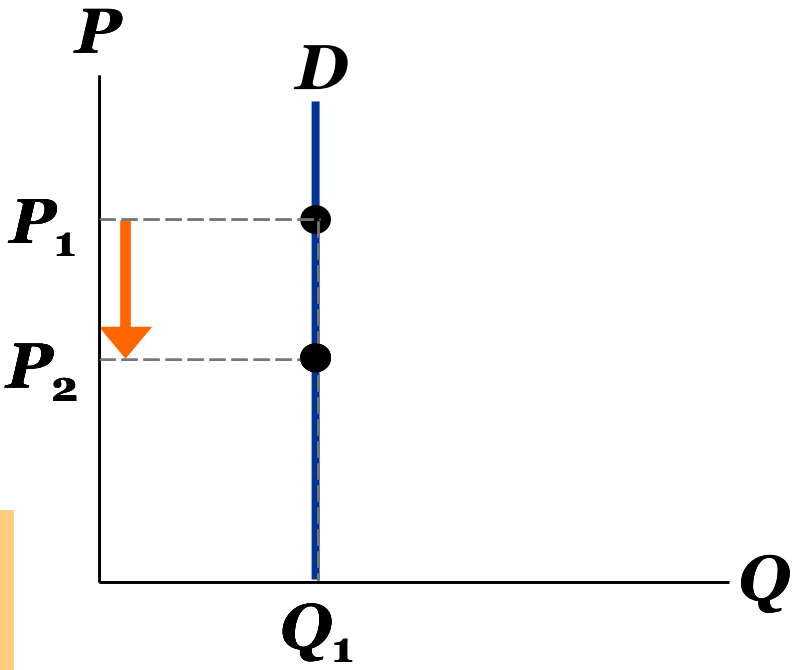
Perfectly inelastic demand

$$\text{Price elasticity of demand} = \frac{\%\ \text{change in } \boldsymbol{Q}}{\%\ \text{change in } \boldsymbol{P}} = \frac{0\%}{10\%} = 0$$

D curve: vertical

Consumers' price sensitivity: none

Elasticity: 0



P falls by 10%

Q changes by 0%

RENMIN BUSINESS SCHOOL

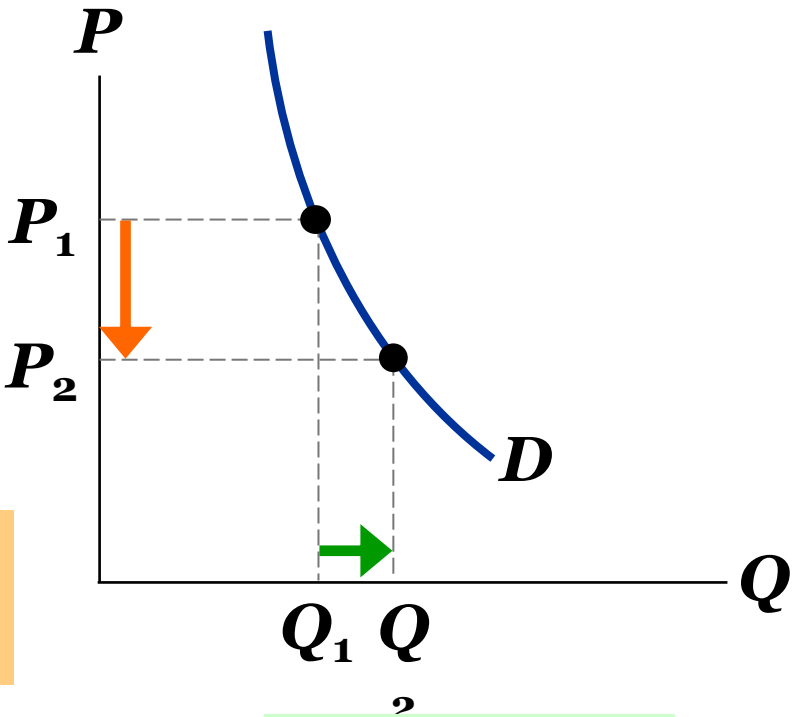
Inelastic demand

$$\text{Price elasticity of demand} = \frac{\%\text{ change in } Q}{\%\text{ change in } P} = \frac{< 10\%}{10\%} < 1$$

D curve:
relatively steep

Consumers' price sensitivity:
relatively low

Elasticity:
< 1



P falls by 10%

Q rises less than 10%

RENMIN BUSINESS SCHOOL

Unit elastic demand

$$\text{Price elasticity of demand} = \frac{\%\text{ change in } Q}{\%\text{ change in } P} = \frac{10\%}{10\%} = 1$$

D curve:

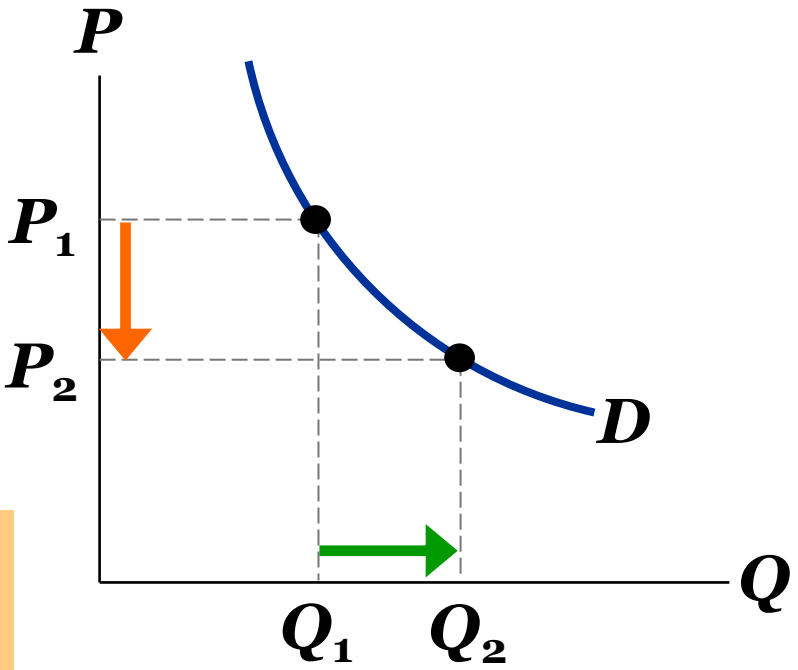
intermediate slope

Consumers' price sensitivity:

intermediate

Elasticity:

1



P falls by 10%

Q rises by 10%

RENMIN BUSINESS SCHOOL

Elastic demand

$$\text{Price elasticity of demand} = \frac{\%\text{ change in } Q}{\%\text{ change in } P} = \frac{> 10\%}{10\%} > 1$$

D curve:

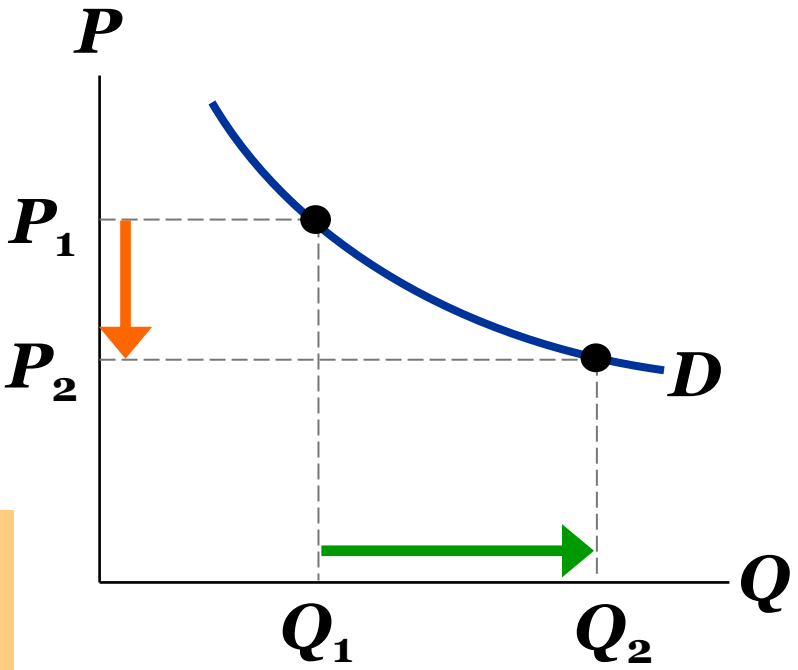
relatively flat

Consumers' price sensitivity:

relatively high

Elasticity:

> 1



P falls by 10%

Q rises more than 10%

RENMIN BUSINESS SCHOOL

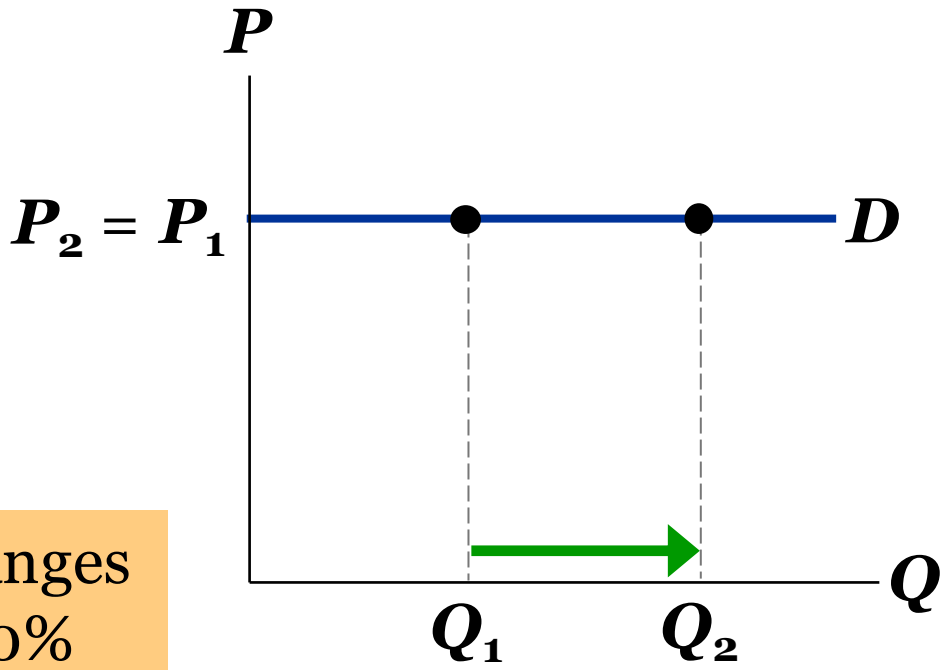
Perfectly elastic demand

$$\text{Price elasticity of demand} = \frac{\%\ \text{change in } \boldsymbol{Q}}{\%\ \text{change in } \boldsymbol{P}} = \frac{\text{any }\%}{0\%} = \text{infinity}$$

D curve:
horizontal

Consumers' price sensitivity:
extreme

Elasticity:
infinity



P changes by 0%

Q changes by any %

RENMIN BUSINESS SCHOOL

Linking Demand Elasticity to Total Revenue

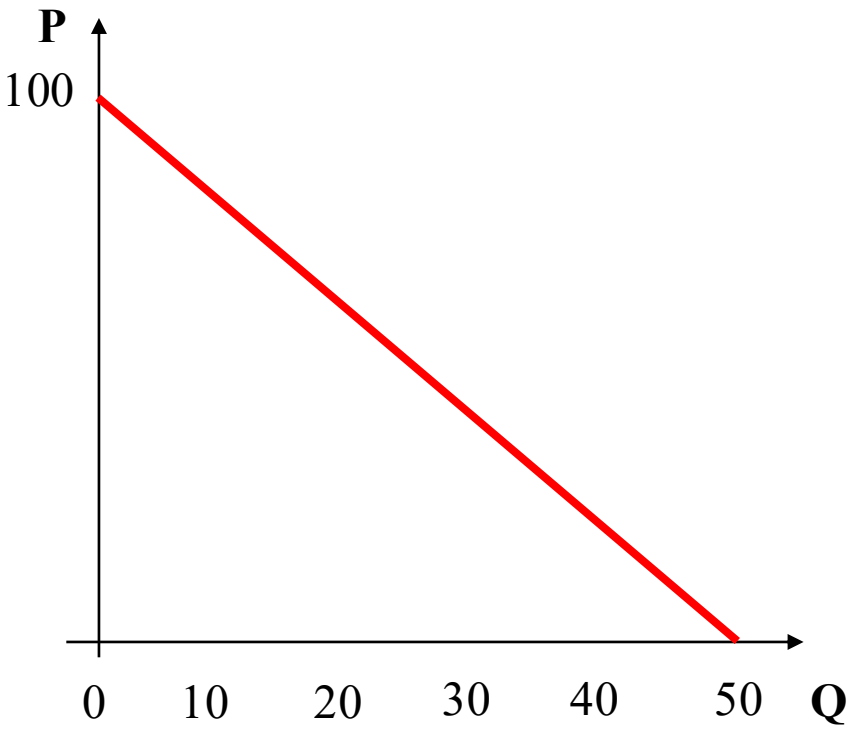
Price Elasticity and Total Revenue

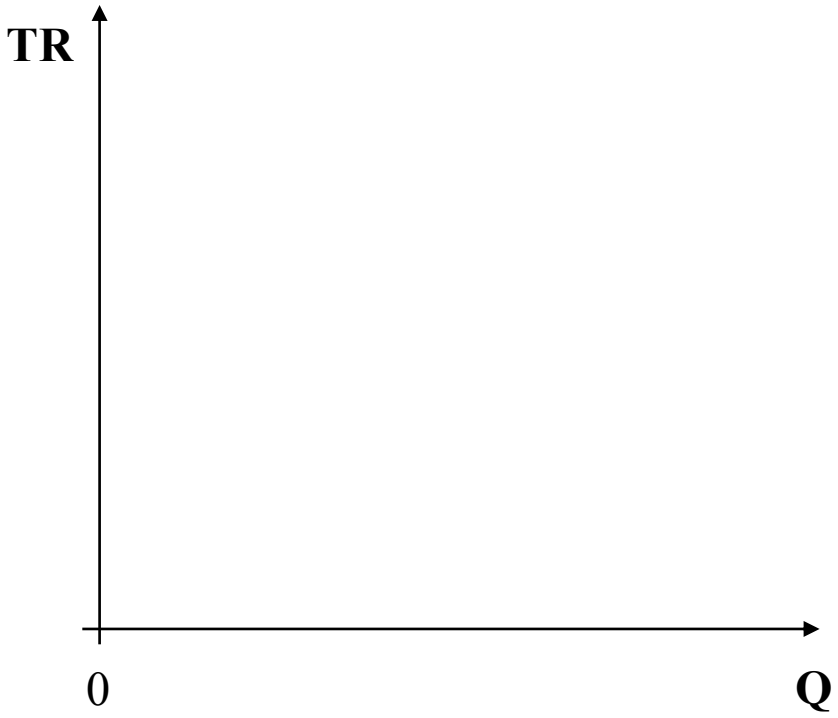
- Continuing our opening discussion, if Bobs raise your price from \$200 to \$250, would your revenue rise or fall?

$$\text{Revenue} = \boldsymbol{P} \times \boldsymbol{Q}$$

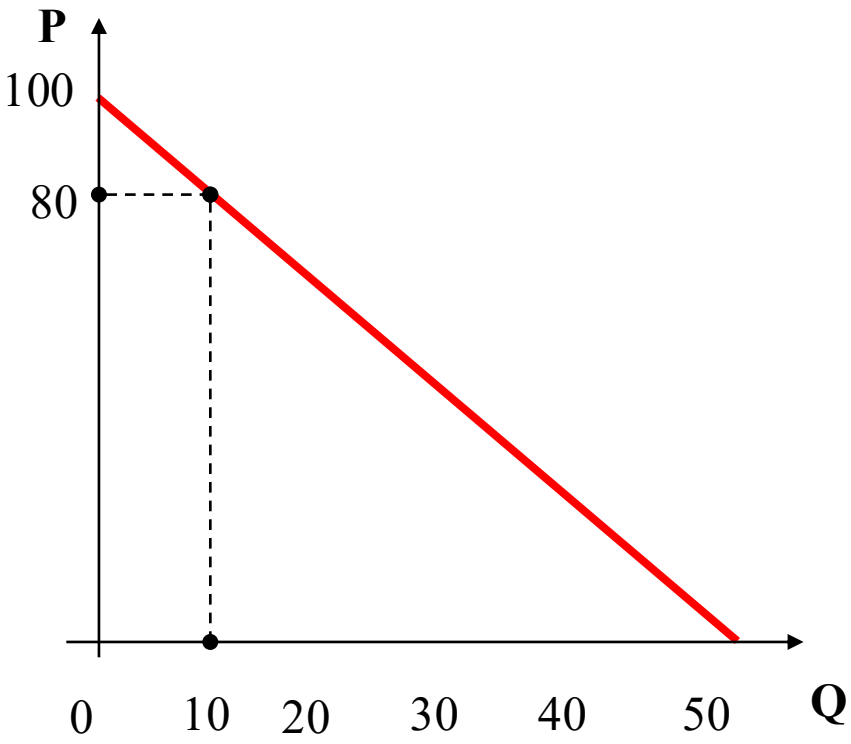
- A price increase has two effects on revenue:
  - Higher ***P*** means more revenue on each unit you sell.
  - But you sell fewer units (lower ***Q***), due to Law of Demand.
- Which of these two effects is bigger?
  It depends on the price elasticity of demand.

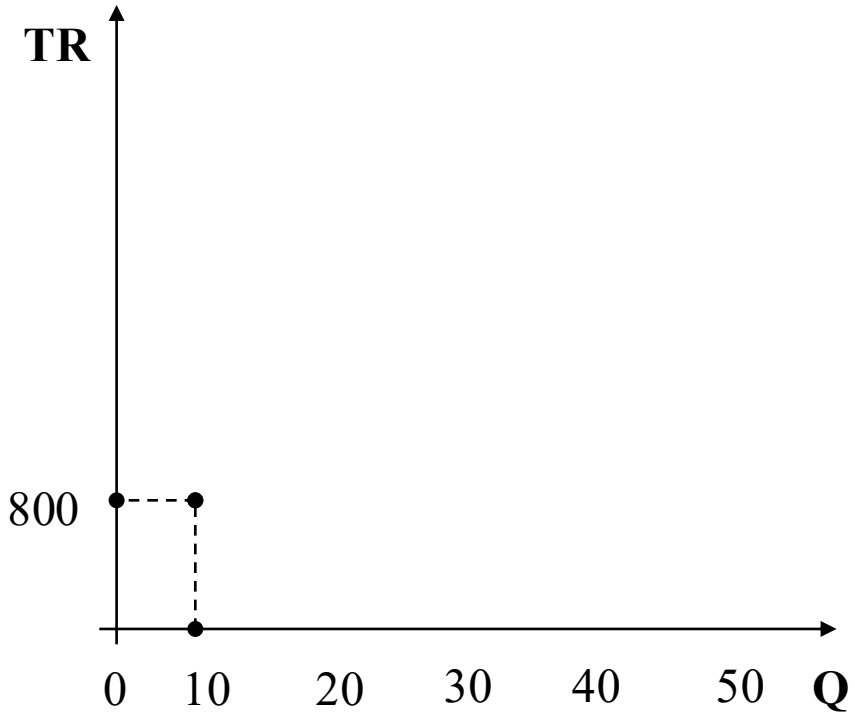
Elasticity and Total Revenue: Linear Demand



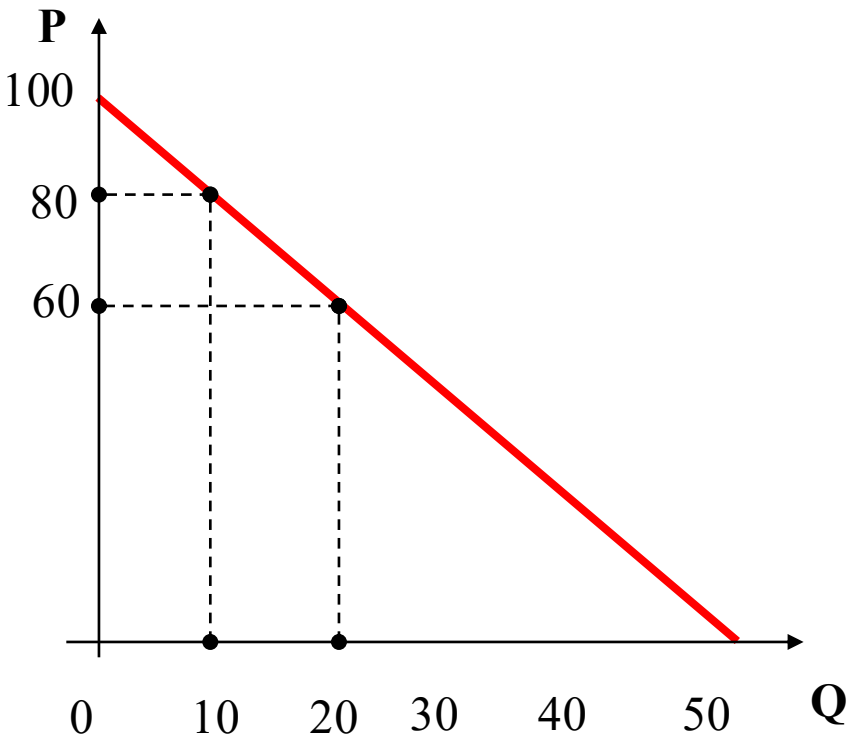


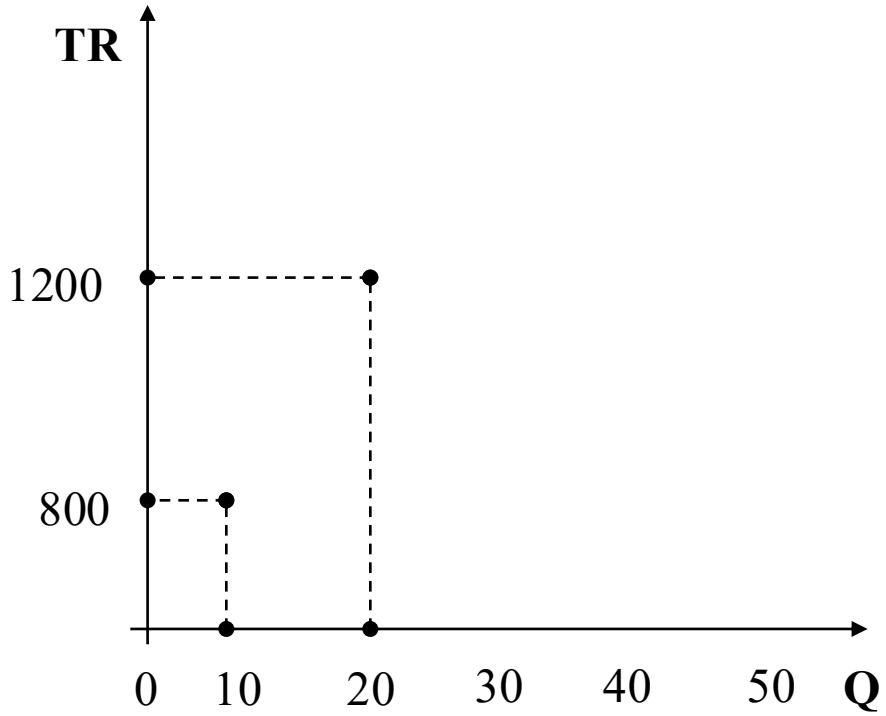
Elasticity and Total Revenue: Linear Demand



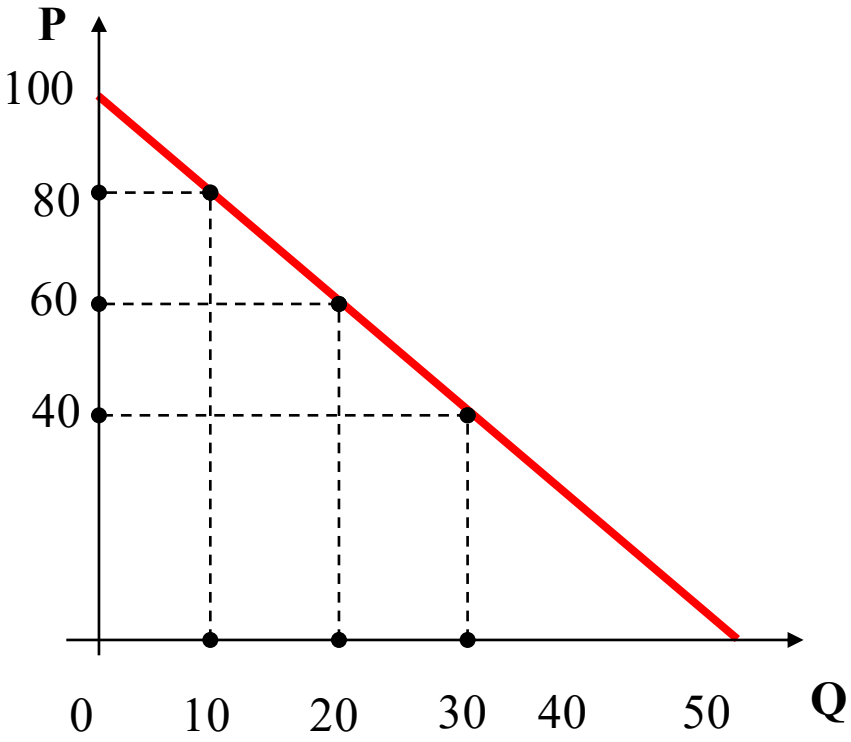


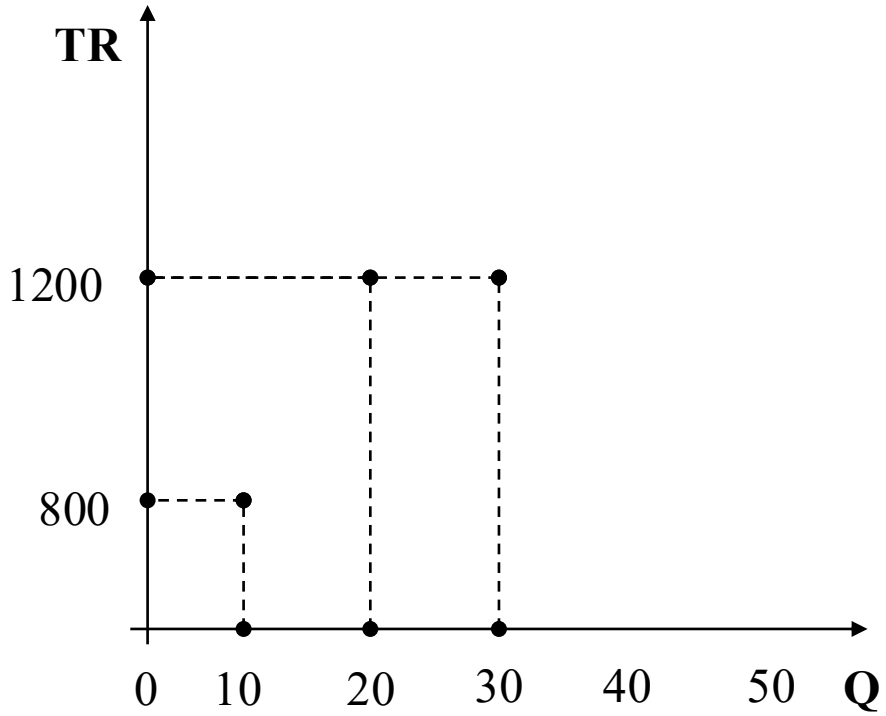
Elasticity and Total Revenue: Linear Demand



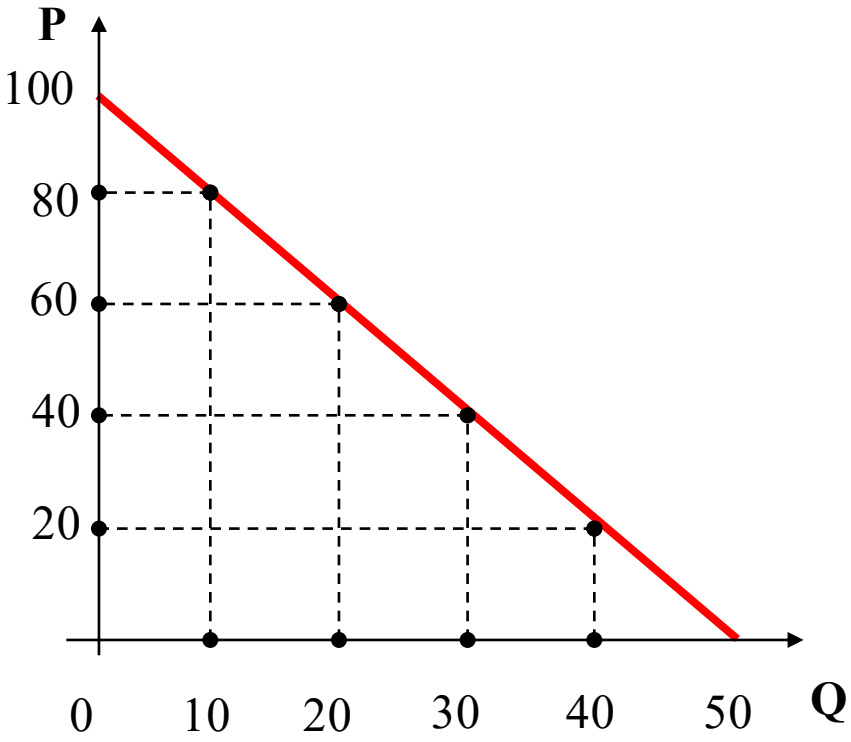


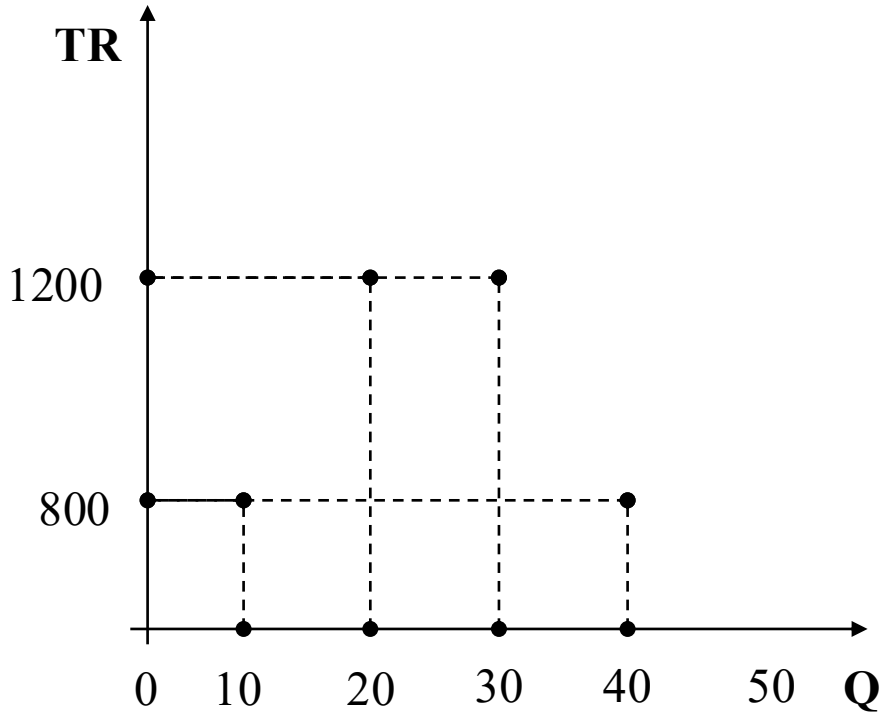
Elasticity and Total Revenue: Linear Demand



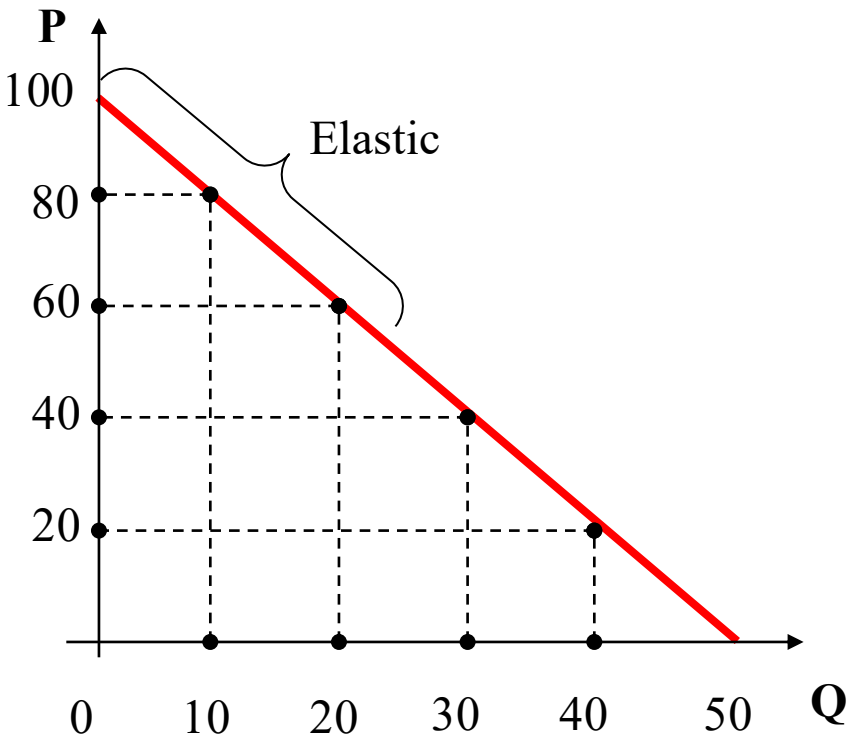


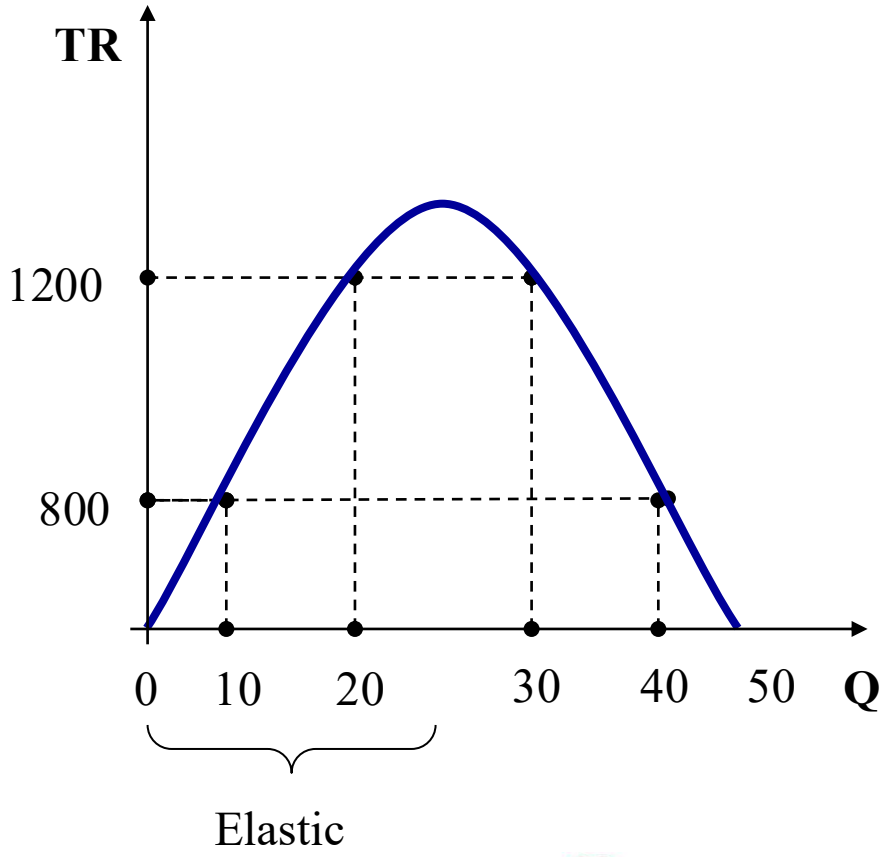
Elasticity and Total Revenue: Linear Demand



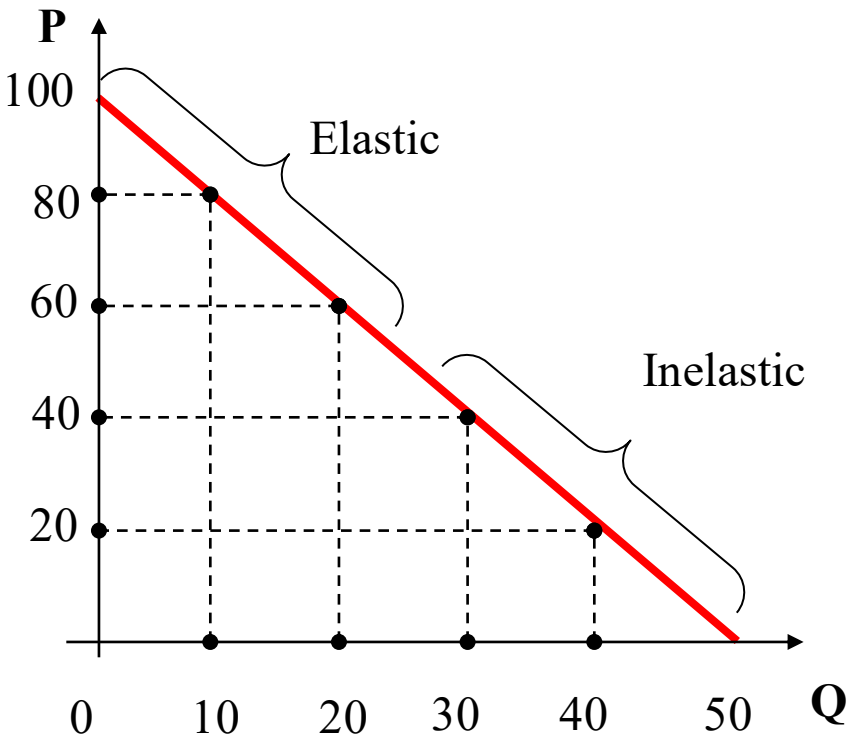


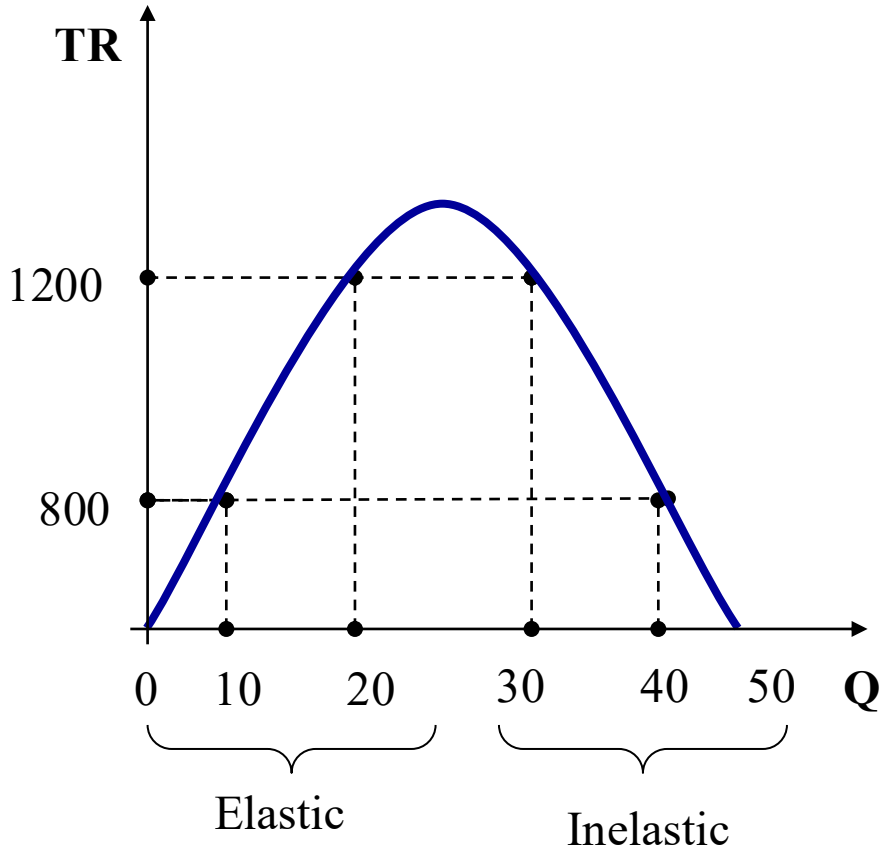
Elasticity and Total Revenue: Linear Demand



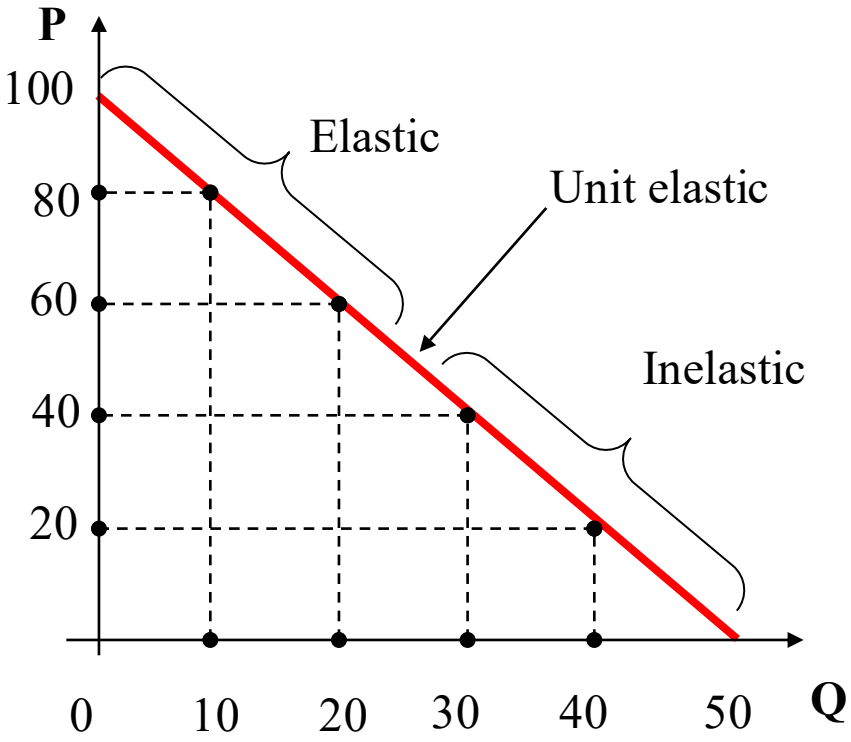


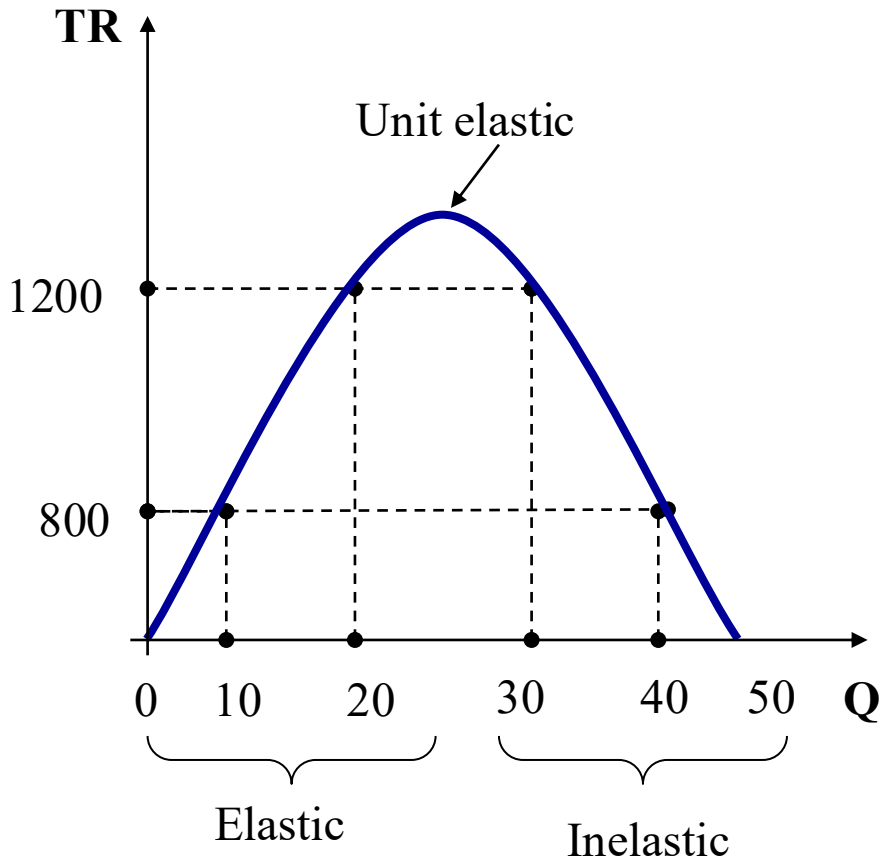
Elasticity and Total Revenue: Linear Demand



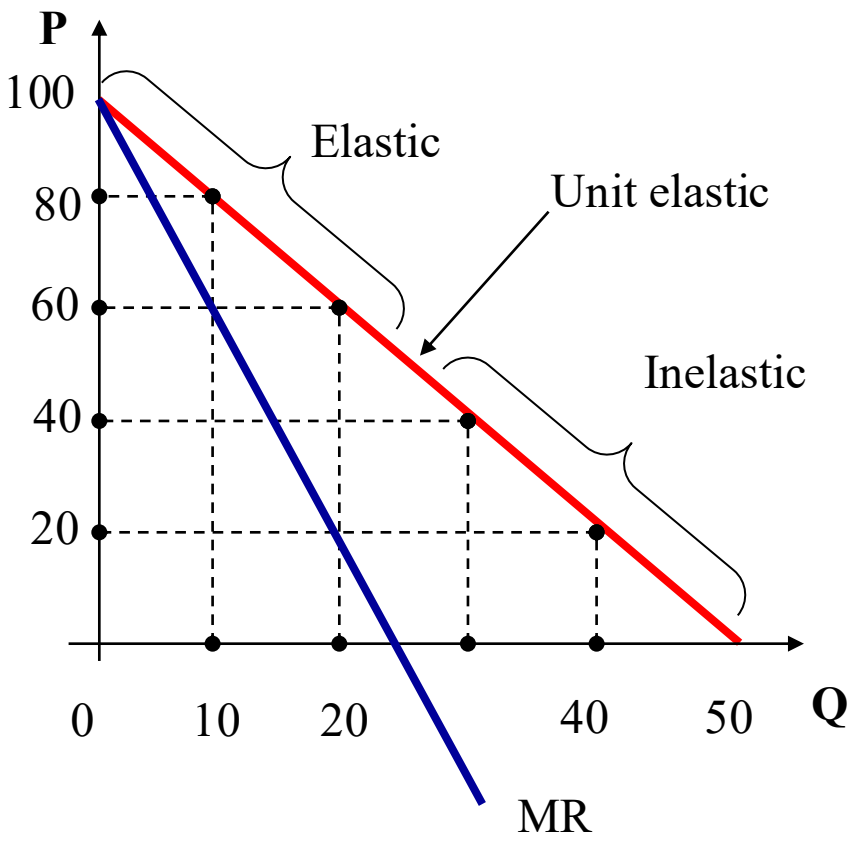


Elasticity and Total Revenue: Linear Demand





Demand, Marginal Revenue (MR) and Elasticity



- For a linear inverse demand function, MR(Q) = a + 2bQ, where b < 0.
- When
  - MR > 0, demand is elastic;
  - MR = 0, demand is unit elastic;
  - MR < 0, demand is inelastic.

RENMIN BUSINESS SCHOOL

Price Elasticity and Total Revenue

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in } \boldsymbol{Q}}{\text{Percentage change in } \boldsymbol{P}} = ?$$

$$\%\text{ change in } \boldsymbol{Q} > \%\text{ change in } \boldsymbol{P}$$

$$\text{Revenue} = \boldsymbol{P} \times \boldsymbol{Q}$$

- The fall in revenue from lower ***Q*** is greater than the increase in revenue from higher ***P***, so revenue falls.

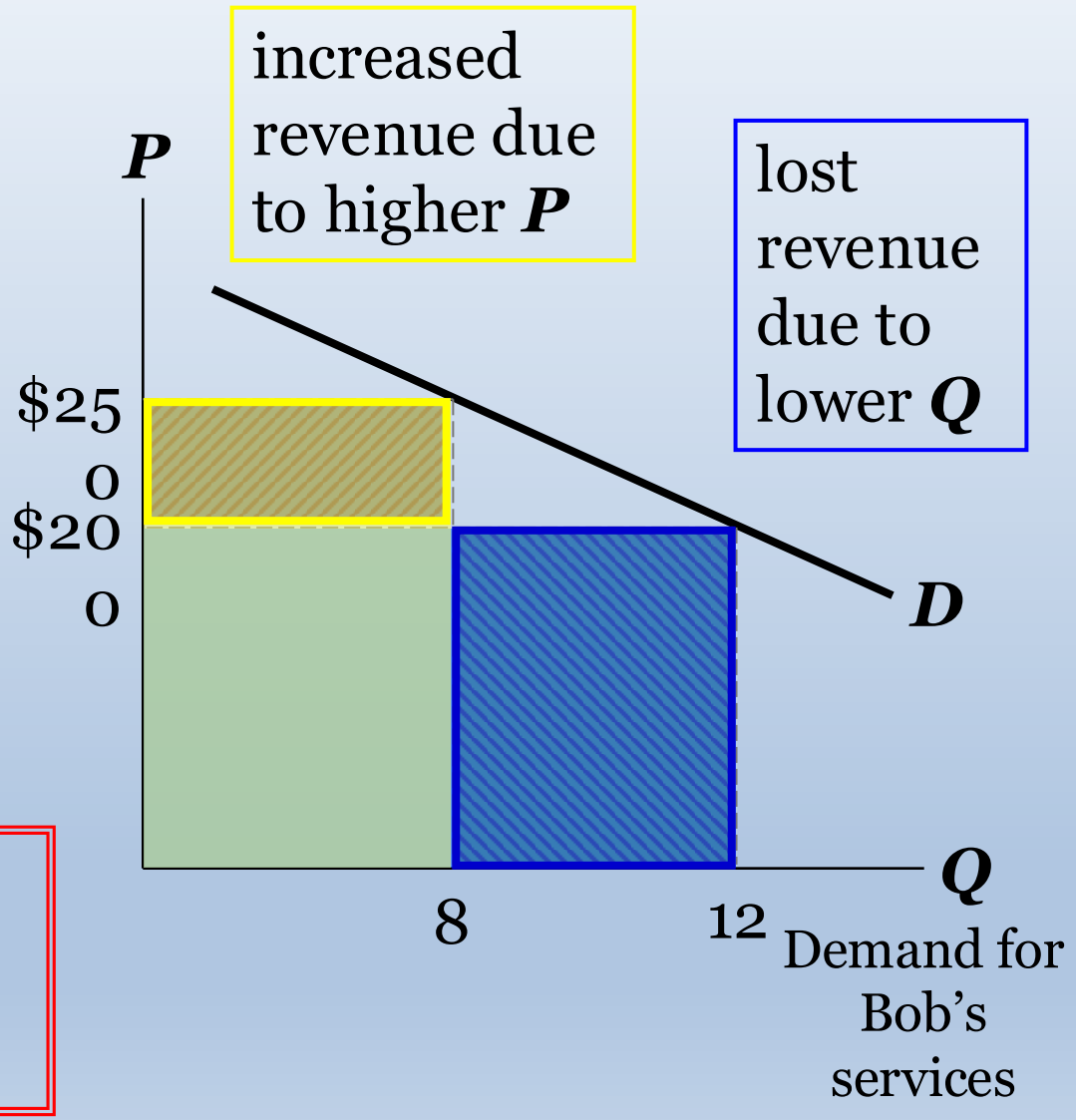
Understand the Change in Bob's Revenue

Elastic demand
(elasticity > 1)

If ***P*** = $200,
Q = 12 and
revenue = $2400.

If ***P*** = $250,
Q = 8 and
revenue = $2000.

When ***D*** is elastic,
a price increase
causes revenue to fall.



Price Elasticity and Total Revenue

- If demand is inelastic, then price elasticity of demand < 1

  % change in ***Q*** < % change in ***P***

Revenue = *P* x *Q*

- The fall in revenue from lower ***Q*** is smaller than the increase in revenue from higher ***P***, so revenue rises.
- In our example, suppose that ***Q*** only falls to 10 (instead of 8) when Bob raises his price to $250.

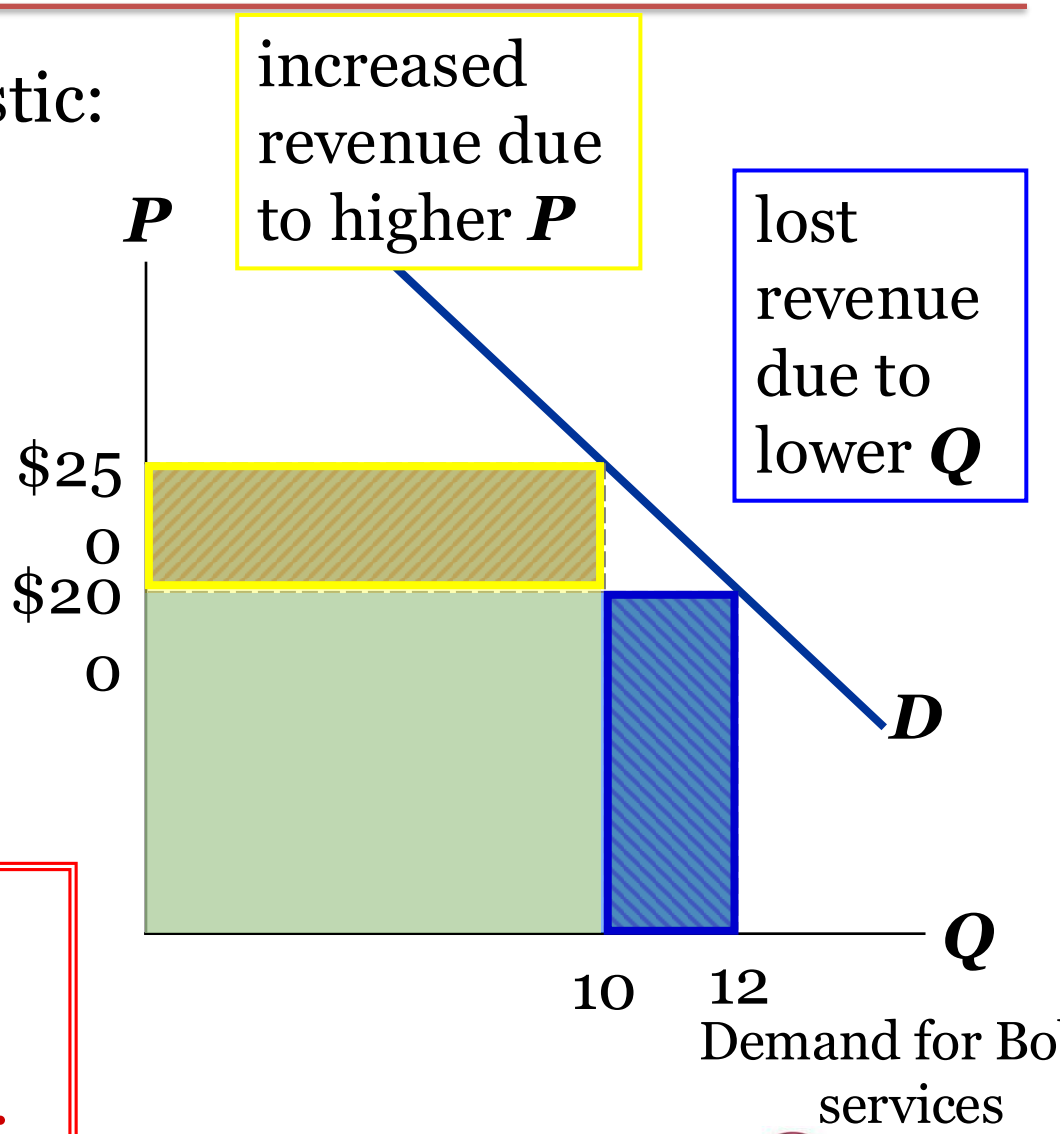
Price Elasticity and Total Revenue

Now, demand is inelastic:
elasticity < 1

If ***P*** = $200,
Q = 12 and
revenue = $2400.

If ***P*** = $250,
Q = 10 and
revenue = $2500.

When ***D*** is inelastic,
a price increase
causes revenue to rise.



RENMIN BUSINESS SCHOOL

Practice I: Elasticity and revenue

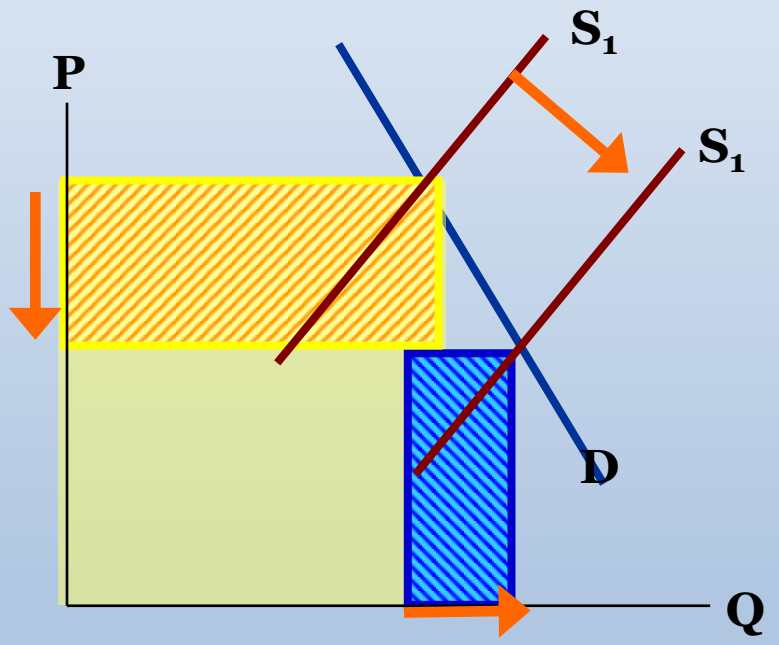
A. Pharmacies raise the price of insulin by 10%. Does total expenditure on insulin rise or fall?

B. As a result of competition, the price of a luxury cruise falls 20%. Does luxury cruise companies' total revenue rise or fall?

Please use diagrams to illustrate them.

Practice II: Elasticity and Expenditure/revenue

Is a harvest year good for peasants? What happens in the long-run?



Other Demand Elasticities

Cross Price Elasticity of Demand

- **Cross Price elasticity of demand（需求的交叉价格弹性）** measures how much Q^d of good X responds to a change in price P of good Y

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in } Q_x^d}{\text{1 Percentage change in } P_y}$$

$$E_{Q_X{}^d, P_Y} = \frac{\%\Delta Q_X{}^d}{\%\Delta P_Y}$$

- If $E_{Q_X^d, P_Y} > 0$, then X and Y are substitutes.
- If $E_{Q_X^d, P_Y} < 0$, then X and Y are complements.

Income Elasticity

- **Income elasticity of demand（需求的收入价格弹性）** measures how much Q^d of good X responds to a change in consumers' income

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in } Q_x^d}{\text{1 Percentage change in } I}$$

$$E_{Q_X,M} = \frac{\%\Delta Q_X{}^d}{\%\Delta I}$$

- If $E_{Q_X,I} > 0$, then X is a normal good.
- If $E_{Q_X,I} < 0$, then X is a inferior good.

Application of Demand Elasticities

Uses of Elasticities

- Elasticities are super useful in managing firms in various ways
  - Pricing.
  - Managing cash flows.
  - Impact of changes in competitors' prices.
  - Impact of economic booms and recessions.
  - Impact of advertising campaigns.
  - And lots more!
- Platforms use big data to design pricing strategies

Classroom Experiment: LLM Membership

- Understand *price elasticity of demand*. 理解需求价格弹性
- Apply it to the **LLM membership market** in China. 应用于中国大模型会员市场
- Analyze implications for *business strategy and policy*. 分析其对企业战略与政策的影响

Notice: You have received a handout about this experiment, please fill out the survey question in the WeChat group. We are going to complete the discussion using the real data collected from you!

Practice: Use of Demand Elasticity

You sell Yangcheng Lake Hairy Crabs in Vanguard Supermarket. Set the price for each Crab

- Customer A: I want ten biggest ones
- Customer B: I want to spend 1,000 Yuan only





Assignment Practice: Pricing and Cash Flows

- According to a report of China's Anti-monopoly Bureau, *China Mobile's* own price elasticity of demand for international long distance services is -10.25.
- *China Mobile* needs to boost revenues in order to meet it's marketing goals.

Question 1: To accomplish this goal, should China Mobile raise or lower it's price?

Quantifying the Change

Question 2: If China Mobile lowered price by 3 percent, what would happen to the volume of international long distance telephone calls routed through China Mobile?

Impact of a change in a competitor's price

- According to a report of China's Anti-monopoly Bureau, China Mobile's (中国移动) cross price elasticity of demand for long distance services is 9.06.

Question 3: If competitors (China Unicom, China Telecom) reduced their prices by 4 percent, what would happen to the demand for China Mobile services?

Summary

- Own Price Elasticity of Demand
  - The sensitive of the change in demand with respect to change in prices
  - Elastic/Unitary/Inelastic demand elasticities
  - Elasticity of Demand characterizes how revenues change when the price changes
- Other elasticities:
  - Cross-price elasticity: Substitutes and Complements
  - Income elasticity
- Concepts of elasticity is super useful in many ways